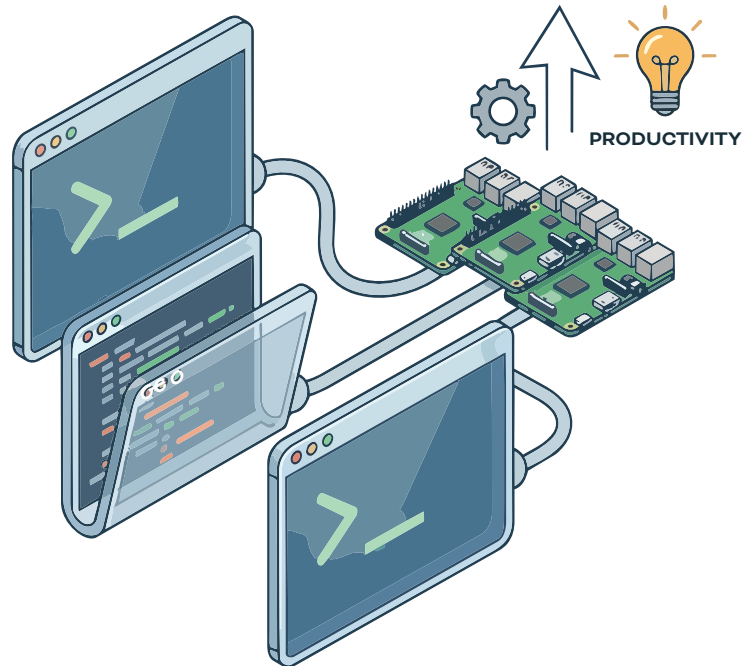


Interactive Shells: Maximising the Productivity of Pi Boards

This hands-on guide will take you through how open source helps to build and deploy user-friendly interactive shells that run on Raspberry Pi Pico boards, making it easier to work with MicroPython.



MicroPython is a popular firmware for programming Raspberry Pi Pico boards but is not easy to work with -- it is not beginner-friendly and lacks flexibility. There's always been a need for a user-friendly interactive shell to play with and develop on Pico boards. There are various free and open source solutions for building and deploying user-friendly shells that run on these boards. The tools presented in this article are tested on an Ubuntu LTS distribution, and we will run them on Docker containers. The Docker engine running on your test machine is the only requirement to try these tools.

Getting started

We'll use various free and open source firmware and/or shell layers to flash the Pico boards. So let's first prepare a few tools required for serial terminal access and file transfer.

First, create the *Dockerfile_MinicomScreen* and build a container image from it using the *docker build -f Dockerfile_MinicomScreen . -t minicomscreen* command:

```
FROM alpine:3.22
```

```
SHELL ["/bin/ash", "-o", "pipefail", "-c"]
WORKDIR /tmp
RUN apk add --no-cache minicom screen \
```

```
&& mkdir /etc/rpipico
```

```
WORKDIR /etc/rpipico
```

Next, create *Dockerfile_Mpremove* and build another container image using the *docker build -f Dockerfile_Mpremove . -t mpremove* command:

```
FROM python:3.14-alpine
```

```
SHELL ["/bin/ash", "-o", "pipefail", "-c"]
WORKDIR /tmp
RUN wget https://github.com/micropython/micropython/archive/
refs/heads/master.zip \
&& unzip master.zip \
&& cp -r micropython-master/tools/mpremote /usr/local/bin/ \
&& rm -rf master.zip micropython-master \
&& pip install -r /usr/local/bin/mpremote/requirements.txt \
&& ln -s /usr/local/bin/mpremote/mpremote.py /usr/local/bin/
mpr \
&& mkdir /etc/rpipico
```

```
WORKDIR /etc/rpipico
ENTRYPOINT ["mpr"]
CMD ["--help"]
```